with fishes such as *Gambusia* which feed on mosquito larvae. Preparations containing *Bacillus thuringiensis* can be sprayed to kill the mosquito larvae since it is not toxic to other forms of life. The best protection against malaria is to avoid being bitten by mosquito. People are advised to use mosquito nets, wire gauging of windows and doors to prevent mosquito bites.

In the 1950's the World Health Organisation (WHO) introduced the Malaria eradication programme. This programme was not successful due to the resistance of *Plasmodium* to the drugs used to treat it and resistance of mosquitoes to DDT and other insecticides.



Malaria vaccine is used to prevent malaria. The only approved vaccine as of 2015 is RTS,S (Mosquirix). It requires four injections and has relatively low efficacy (26–50%). Due to this low efficacy, WHO does not recommend the use of RTS,S vaccine in babies between 6 and 12 weeks of age.

7.1.3 Fungal diseases

Fungi was recognized as a causative agent of human diseases much earlier than bacteria. Dermatomycosis is a cutaneous infection caused by fungi belonging to the genera *Trichophyton*, *Microsporum* and *Epidermophyton*.

Ringworm is one of the most common fungal disease in humans **(Fig. 7.4)**. Appearance of dry, scaly lesions on the skin, nails and scalp are the main symptoms of the disease. Heat and moisture help these fungi to grow and makes them to thrive in skin folds such as those in the groin or between the toes. Ringworms of the feet is known as **Athlete's foot** caused by *Tinea pedis* **(Fig. 7.5)**. Ringworms are generally acquired from soil or by using clothes, towels and comb used by infected persons.



Fig. 7.4 Symptoms of ringworm



Fig. 7.5 Symptoms of Athlete's foot

7.1.4 Helminthic diseases

Helminthes are mostly endoparasitic in the gut and blood of human beings and cause diseases called **helminthiasis.** The two most prevalent helminthic diseases are Ascariasis and Filariasis.

Ascaris is a monogenic parasite and exhibits sexual dimorphism. **Ascariasis** is a disease caused by the intestinal endoparasite *Ascaris lumbricoides* commonly called the **round worms (Fig. 7.6)**. It is transmitted through ingestion of embryonated eggs through contaminated food and water. Children playing in contaminated soils are also prone to have a chance of transfer of eggs from hand to mouth. The symptoms of the disease are abdominal pain, vomiting, headache, anaemia, irritability

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